IVI version 2.0
Guideline document

Version number: 1.1
Version date: 2025-05-11

Revision history

VersionNr.	VersionDate	Author(s)	Description
1.1	2025-05-11	P. Oude Weernink	Option to use a Seal for signing added.
1.0	2025-01-19	P. Oude Weernink	Final version.
0.99	2024-11-17	P. Oude Weernink	Several text corrections and additional information added.
0.91	2024-10-15	P. Oude Weernink	Several text corrections and additional information added.
0.9	2024-08-24	P. Oude Weernink	Several text corrections and additional information added.
0.6	25-4-2024	P. Oude Weernink	Usage of Range values and Minimum/Maximum values added
0.5	09-04-2024	P. Oude Weernink	Several remarks added
0.4	12-02-2024	P. Oude Weernink	Several remarks added
0.3	14-01-2023	P. Oude Weernink	Remarks KBA How to deal with corrections
0.2	10-11-2023	P. Oude Weernink	Remarks NorType
0.1	31-10-2023	P. Oude Weernink	Initial draft version

Content

1. General principles

1.1 Signature

In version 2.0 a digital signature will be made mandatory.
Version 1.9: <xs:element ref="sig:Signature" minOccurs="0" maxOccurs="1"/>
Version 2.0: <xs:element ref="sig:Signature"/>

Each eCoC file has to be signed according to:
2024-1061: **Article 1 Requirements for secure data exchange**

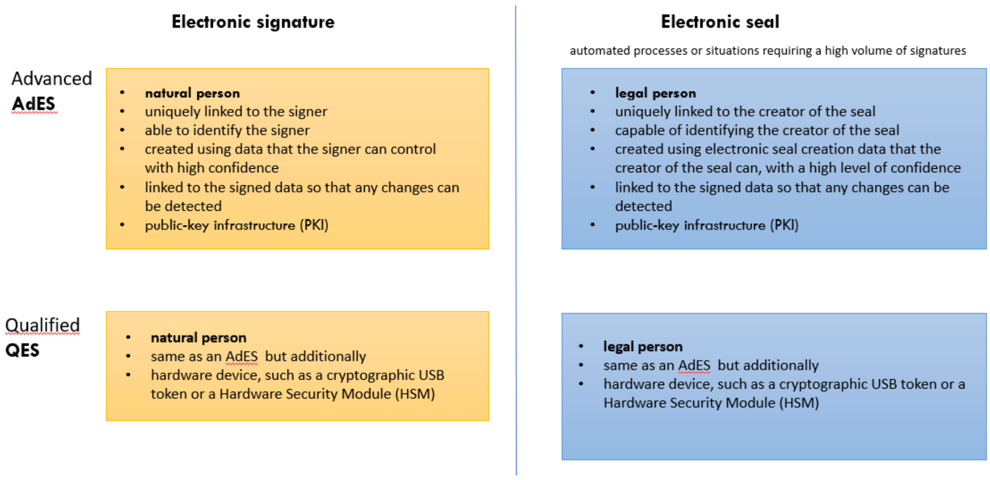
3. The exchange of messages between manufacturers and national access points of EUCARIS and within EUCARIS shall be
secured using an electronic signature within the meaning of Regulation (EU) No 910/2014 and data encryption protocols.
4. The national access points of EUCARIS and the EUCARIS nominated body for operation shall provide for validation
of the electronic signatures by verifying and confirming that the requirements provided for in Article 26 of Regulation

(EU) No 910/2014 were met at the time of signing.
Regulation (EU) No 910/2014 (hereafter the eIDAS Regulation or eIDAS),
on electronic identification and trust services for electronic transactions in the internal market,
provides a regulatory environment for electronic identification of natural and legal persons and for trust services.

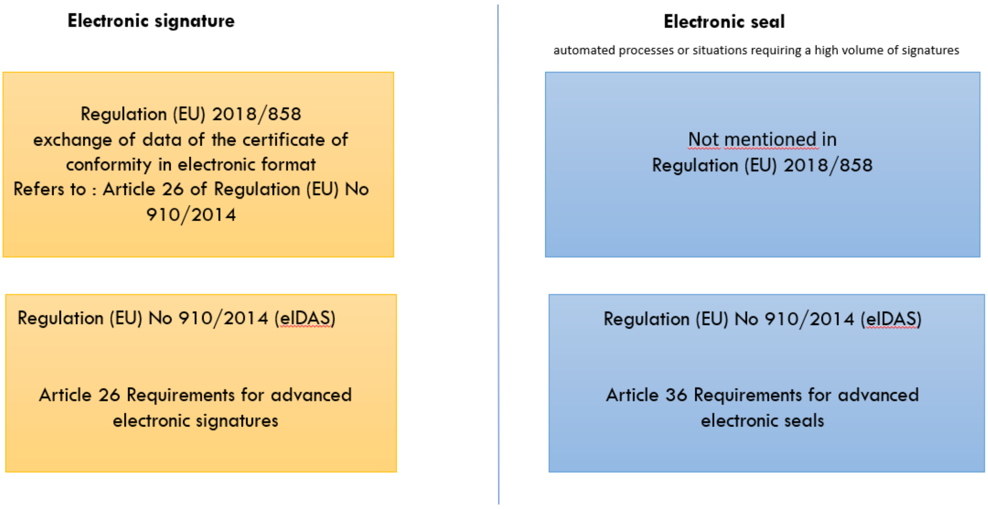
Article 26 Requirements for advanced electronic signatures
In general, using an advanced signature is easier and more cost efficient then using qualified signatures. For IVI the requirement is to use an advanced signature which is in compliance with the eIDAS regulation. Also a qualified signature is allowed to use.

Through the eIDAS dashboard a list of Trusted Service Provider is available: [eIDAS Dashboard (europa.eu)]

Note:
although not mentioned in the implementing act it has been decided that next to a electronic signature also a Seal will be allowed.



	Electronic signature	Electronic seal automated processes or situations requiring a high volume of signatures
Advanced **AdES**	• **natural person** • uniquely linked to the signer • able to identify the signer • created using data that the signer can control with high confidence • linked to the signed data so that any changes can be detected • public-key infrastructure (PKI)	• **legal person** • uniquely linked to the creator of the seal • capable of identifying the creator of the seal • created using electronic seal creation data that the creator of the seal can, with a high level of confidence • linked to the signed data so that any changes can be detected • public-key infrastructure (PKI)
Qualified **QES**	• **natural person** • same as an AdES but additionally • hardware device, such as a cryptographic USB token or a Hardware Security Module (HSM)	• **legal person** • same as an AdES but additionally • hardware device, such as a cryptographic USB token or a Hardware Security Module (HSM)



With effect from: 5th July 2026: it's mandatory to sign all IVI files using at least eIDAS compliant advanced seals certificates.

The eCoC file can only be signed by :

The manufacturer
The EU representative of the manufacturer
Both parties have to be compliant and stated in the WVTA.

1.2 New fields should only be added after publication of the regulation

The consequence of this will probably be that the time to implement the changes will be too short. Until a process with the Commission (Forum) is in place the preparation of the IVI file has to be ready before the actual publication. Until a new update process is in place we will maintain the current process.

1.3 Usage of Remarks

In some cases regulations refer to adding information in the Remarks field before the CoC regulations has been changed an de new field has been added there. A current example is "Additional mass due to batteries: …… kg" (*1) mentioned in 2024/883.

In these cases the field can already be added to the IVI version before the actual legislation is in place. However it cannot be mandatory for the manufacturer to use this field instead of the Remarks before the actual CoC legislation has been updated.

1.4 Additional fields not part of CoC dataset

No fields should be entered which are not part of the CoC dataset for the specified vehicle category. For instance do not enter Colour for the vehicle category M2.

1.5 Restrict fields to latest Regulations and onwards

Limit the fields in IVI 2.0 to the fields mentioned in :

- 2018/858
- 167/2013

- 168/2013

Consequence:

CoC's for vehicles produced on or after July 5th 2026 have to be send in using IVI version 2.0. This implies that if vehicles which have been produced after this date using other Type approval numbers then the three mentioned above, the CoC data in the IVI message also has to comply with IVI version 2.0.

1.6 Fields not mentioned in CoC related regulations will not be part of IVI 2.0

Consequences:

- Removal of TechnicalDataGroup.
- Removal of NationalDataGroup.
- Removal of Individual Approval Certificate fields.
- Removal of CoC fields which are not in use anymore.
- Removal of CoC fields for other framework directives then mentioned.

1.7 All CoC files shall contain a version number

The field will be mandatory in version 2.0.
Version 1.9: <xs:element name="VersionNumberXsd" minOccurs="0" type="VersionNumberXsd"/>
Version 2.0: <xs:element name="VersionNumberXsd" type="VersionNumberXsd"/>

- The first IVI message published shall always contain VersionCoC = 1
- Corrections on this IVI message will have a subsequent version number

1.8 Backwards compatible

Within the main versions (1.0 – 2.0 etc.) IVI shall be backwards compatible.
Consequence:

- A new enumeration for an already existing field breaks the backwards compatibility. This means enumerations have to be added immediately when adding new fields.

1.9 Check IVI version related to DateOfManufacture

All IVI files with a DateManufacture after July 5th 2026 have to comply to version 2.0. This will also be the case for the 168/2013 and 167/2013 IVI files, for these guidelines this cannot be checked as the DateManufacture is not part of the CoC dataset in this case.

1.10 Limitation in time

Older versions of the IVI will always be relevant:

Sending in corrections in the same version as the original version;
For purpose of re-registration (if the vehicle hasn't been modified).

It's not possible to set a limitation in time on accepting previous IVI versions.

1.11 Move non CoC fields to header

Fields which are not part of the CoC file but needed as meta data will be part of the Header group.

IviReferenceId
IviVersionNumberXsd
IviVersionNumber
IviVersionDateTime
IntendedCountryRegistration
DesignatedTypeApprovalCountry

What we need to know to check versioning and make error handling easier:

- A unique reference to the XML file (**IVIReferenceId**).
  Reason: technical purposes for unique referencing.

- Which version of the XSD has been used to generate the XML file (**IviVersionNumberXsd**).
  Reason: handle the ICM correctly and to check the XML against the correct XSD file.
- If the CoC XML file is a correction of a previously send in version(**IviVersionNumber**).
  Reason: corrections are handled differently in the registration process.
- Date and Time of the issuing of the IVI message (**IviVersionDateTime**).
  Reason: technical purposes and error handling.

1.12 Corrections of IVI messages

It’s possible to send a corrected version of an IVI xml message. A correction always has to be send in using the same IVI XSD version as the original version.
General rules for handling corrections:

- the **IVIReferenceId** will have a new Id
- the **IviVersionNumberXsd** will be the same as the original version
- the **IviVersionNumber** will have to be the previous version + 1
- the **IviVersionDateTime** has to be changed to the date the correction has been set up

1.13 How to use Indicator fields

In case of an indicator field or a field with a code that indicates not applicable (i.e None), the field must always be included in the eCoC xml when it is mentioned in the CoC regulations. This will be checked in the new version of the ICM.

1.14 Field types definitions

Field Type	**Definition**
Text	Text and the length behind the element states the possible maximum length for the specified attribute.
Date	Stated as: YYYY-MM-DD (2023-08-13)
DateTime	Stated as: 2015-03-17T09:55:21.351+01:00
..Code	Enumeration of Codes which limits the data entered in the tag.
Indicator	Can have value Y or N. If not relevant no tag to be added in the XML file.
Decimal32	Three numbers in total, 2 positions behind the separator. (9.99)
Decimal52	Five numbers in total, 2 positions behind the separator. (999.99)
Decimal53	Five numbers in total, 3 positions behind the separator. (99.999)
Decimal62	Six numbers in total, 2 positions behind the separator. (9999.99)
Decimal63	Six numbers in total, 3 positions behind the separator. (999.999)
Decimal64	Six numbers in total, 4 positions behind the separator. (99.9999)
Decimal72	Seven numbers in total, 2 positions behind the separator. (99999.99)
Decimal75	Seven numbers in total, 5 positions behind the separator. (99.99999)
Decimal82	Eight numbers in total, 2 positions behind the separator. (999999.99)
Decimal86	Eight numbers in total, 6 positions behind the separator. (99.999999)
Decimal95	Nine numbers in total, 5 positions behind the separator. (9999.99999)
Decimal106	Four numbers in total, 6 positions behind the separator. (9999.999999)
Integer xs:nonNegativeInteger	States that the field has to contain a positive numeric value. The length behind the element states the length for the numeric value.
Negative values	If negative values should be possible, a field type to allow the entry for these values has been added. For instance: <xs:simpleType name="RoadLoadCoef"> <xs:restriction base="xs:decimal"> <xs:totalDigits value="9"/> <xs:fractionDigits value="6"/> <xs:minInclusive value="-999.999999"/> <xs:maxInclusive value="999.999999"/> </xs:restriction>

	</xs:simpleType>

2. Naming conventions

2.1 Introduction

The current version of the Initial Vehicle Information (IVI) message has been set up and implemented as of 1st of January 2016. During the years up until the current version a lot of changes have been implemented. These changes have been implemented without guidelines on naming conventions of the tags within the IVI message. As the set-up of a completely new version, which will not be backwards compatible anymore, now is the time to set up guidelines for naming conventions to make sure future versions can be set up with these guidelines in mind.
This document describes the guidelines for the definition of new tag names and will have an impact on current tag names and grouping of attributes.

2.2 Keep tag names consistent

Make sure the tag name can be easily related to the naming in the regulation. If in different regulations different names are used select the name which is the most meaningful or add additional information to the tag name.
For instance if for some vehicle category the field types are different use the vehicle definition in the tag name: LocationStatutoryPlate23Wheel.

2.3 Keep tag names unique

Do not use the same tag names within the IVI message. These should always be unique even when they occur in separate groups or tables.

2.4 Usage of Range values and Minimum/Maximum values

Always use “Min”/”Max“ if the value is a range value

example: AllowedActualMass
AllowedActualMassMin
AllowedActualMassMax

Always use “Adjustable” if the value is a min/max and can be adjusted

example: AxleTrack
AxleTrackAdjustableMin
AxleTrackAdjustableMax

Always use minimum of maximum if this is the exact value

example: AxleSpacing
AxleSpacingMinimum
AxleSpacingMaximum

Use “Variable” in stead of “Adjustable” for Masses which aren’t adjustable:

example: MassInRunningOrderVariableMin

2.5 Table- and Group names

The table - and group names may contain more context then the underlying tag names. However the tag names should be self-explanatory.

Example:

```
<xs:complexType name="TestWltpCO2Group">
 <xs:all>
  <xs:element name="WltpCo2Low" minOccurs="0" type="Decimal52"/>
  <xs:element name="WltpCo2Medium" minOccurs="0" type="Decimal52"/>
  <xs:element name="WltpCo2High" minOccurs="0" type="Decimal52"/>
  <xs:element name="WltpCo2ExtraHigh" minOccurs="0" type="Decimal52"/>
  <xs:element name="WltpCo2Combined" minOccurs="0" type="Decimal52"/>
```

```
  <xs:element name="WltpCo2WeightedCombined" minOccurs="0" type="Decimal52"/>
 </xs:all>
</xs:complexType>
```

2.6 Keep tag names limited to 128 character

The limit to the length of a tag name is set to 128 characters.

2.7 Pascal Case start with Capital

Use Pascal Case starting with a Capital. Within the tag names switch to Capitals when it's not one whole word anymore.

2.8 Usage of symbols.

Never use symbols in the structure or in the element names. Only use letters or numbers.

2.9 Standardise abbreviations

If abbreviations have to be used, always use the same abbreviation in the tag names.
Example: For Vehicle always use Veh if an abbreviation is needed. Avoid abbreviations if possible, and do not use abbreviations in Table or Group names.

2.10 Units

Other units for already existing fields
When for an existing field the unit changes, always add the unit to the tag name for the changed field.

2.11 Use domain types

In the current version every field has its own type definition. This will be optimized by using more general type definitions where possible. An example is:
Version 1.9: <xs:element name="WltpType1Co" minOccurs="0" type=" WltpType1Co"/>
Version 2.0: <xs:element name="WltpType1Co" minOccurs="0" type=" Decimal82"/>

The fields with standard enumerations will still have a unique type definition:
<xs:element name="VehicleClass" minOccurs="0" type="VehicleClassCode"/>

2.12 Re-use of outdated element names.

When an element becomes outdated it should not be possible to re-use the name later for another meaning. For example the re-use of MaximumNetPowerElectricEngine and MaximumContinuousRatedPowerVeh (consolidated power).

2.13 Harmonization between base directives.

It is important to harmonize the element names between the base directives. Try to use the same elements for all category vehicles if the information is the same.